BYOP Project Report

Student Grade Predictor

Predicting Final Scores & Letter Grades using Machine Learning

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Models	Linear Regression, Decision Tree

1. Introduction

Predicting academic performance is a classic and highly relevant problem in the field of educational data mining. Students, educators, and institutions all benefit from having early indicators of how a student is likely to perform at the end of a semester. Early intervention based on such predictions can significantly improve student outcomes.

This project, developed as part of the Bring Your Own Project (BYOP) assignment for the Fundamentals of AI and ML course, builds a Student Grade Predictor using two supervised learning models — Linear Regression for numeric score prediction and a Decision Tree Classifier for letter grade prediction.

1.1 Motivation

The motivation behind choosing this problem is its direct real-world applicability. During coursework, it is common to see students who are uncertain about their academic standing. A simple ML-powered tool that takes in basic, easily obtainable inputs — like attendance, study hours, and past performance — and returns a predicted score and grade can serve as a useful self-assessment tool for students.

1.2 Objectives

- Build a Linear Regression model to predict a student's final exam score
- Build a Decision Tree Classifier to predict the letter grade (A, B, C, D, or F)
- Visualize feature relationships and model performance
- Interpret which factors most strongly influence academic outcomes

2. Dataset

Since no suitable real-world dataset was readily available with the exact combination of features required, a synthetic dataset of 300 student records was generated programmatically using NumPy. The dataset is realistic in its structure and relationships, with controlled noise added to simulate real-world variability.

2.1 Features Used

Feature	Description	Range
attendance_pct	Percentage of classes attended	50 – 100
study_hours_day	Average daily study hours	1 – 10 hrs
prev_exam_score	Score in the previous exam	40 – 100
assignments_done	Number of assignments submitted (out of 10)	0 – 10
sleep_hours	Average nightly sleep in hours	4 – 9 hrs

2.2 Target Variables

- final_score — A continuous value from 0 to 100, used as the regression target
- grade — A categorical label (A, B, C, D, F) derived from the final score, used as the classification target

2.3 Grade Mapping

Score Range	Grade
90 – 100	A
75 – 89	B
60 – 74	C
45 – 59	D
Below 45	F

3. Approach and Methodology

The project follows a standard supervised machine learning pipeline. The steps are described below.

3.1 Data Generation

A synthetic dataset was created using NumPy's random number generators. The final score was computed as a weighted linear combination of input features with added Gaussian noise, ensuring a realistic score distribution while maintaining logical relationships between features.

3.2 Exploratory Data Analysis

Before training models, the dataset was explored visually through histograms of each feature and a correlation heatmap. Key findings from EDA included a strong positive correlation between study_hours_day and final_score, and a moderate positive correlation between attendance_pct and final_score.

3.3 Model A — Linear Regression

Linear Regression was chosen to predict the continuous final score. It fits a linear equation to the training data, with each feature assigned a coefficient reflecting its contribution to the predicted score. The model was trained on 80% of the data and evaluated on the remaining 20%.

3.4 Model B — Decision Tree Classifier

A Decision Tree Classifier was trained to predict the letter grade. The tree was limited to a maximum depth of 5 to prevent overfitting while maintaining interpretability. The same train/test split was used for fair comparison.

3.5 Evaluation Metrics

Model	Metric	Description
Linear Regression	MAE	Mean Absolute Error — average prediction error
Linear Regression	RMSE	Root Mean Squared Error — penalises large errors
Linear Regression	R^2 Score	Proportion of variance explained by the model
Decision Tree	Accuracy	Fraction of grades correctly classified
Decision Tree	Confusion Matrix	Breakdown of correct and incorrect predictions by class

4. Results

4.1 Linear Regression Performance

Metric	Value
Mean Absolute Error (MAE)	3.90
Root Mean Squared Error	4.77
R^2 Score	0.8739

The R^2 score of 0.87 indicates that the model explains approximately 87% of the variance in student scores — a strong result for a beginner-level project with synthetic data.

4.2 Feature Coefficients

Feature	Coefficient
study_hours_day	+4.17
sleep_hours	+0.98
assignments_done	+1.29
attendance_pct	+0.28
prev_exam_score	+0.25

Study hours has the highest coefficient (+4.17), meaning every additional hour of daily study adds approximately 4 points to the predicted final score.

4.3 Decision Tree Performance

The Decision Tree Classifier achieved an accuracy of 53.33% on the test set. While this may seem modest, it reflects the difficulty of classification near grade boundaries (e.g. a score of 74.9 vs 75.0 determines B vs C). Improving this would require more data or additional features.

4.4 Sample Prediction

Input	Value
Attendance	85%
Study Hours/Day	6.0 hrs
Previous Score	72
Assignments Done	8 / 10
Sleep Hours	7.0 hrs
Predicted Score	88.0 / 100
Predicted Grade	B

5. Challenges Faced

- Grade classification near boundaries: Students with scores like 74 vs 75 are classified very differently (C vs B). The model struggles at these boundary cases, which is expected.
- Generating realistic synthetic data: Ensuring the dataset reflected plausible real-world patterns while keeping relationships learnable required careful tuning of the score formula.
- Class imbalance: Very few F-grade students were generated (2 out of 300), which caused the classifier to rarely predict F. In future work, techniques like SMOTE could address this.

6. What I Learned

- How to apply Linear Regression end-to-end: from data preparation to coefficient interpretation
- How Decision Trees make splits based on feature thresholds and how depth affects overfitting
- The importance of EDA — visualising correlations before modelling revealed which features matter most
- How to evaluate regression and classification models using appropriate metrics
- How synthetic data generation works and how to control relationships between variables

7. Conclusion

The Student Grade Predictor successfully demonstrates the application of supervised machine learning to a real-world educational problem. The Linear Regression model performs with high accuracy (R^2 = 0.87), while the Decision Tree provides an interpretable classification into letter grades. The most significant finding is that daily study hours is the strongest predictor of academic performance — a result that aligns well with common understanding.

The project, though built on synthetic data, establishes a solid foundation that could be extended with real institutional data, a web interface, or additional features like subject-wise performance, extracurricular activities, and mental health indicators.